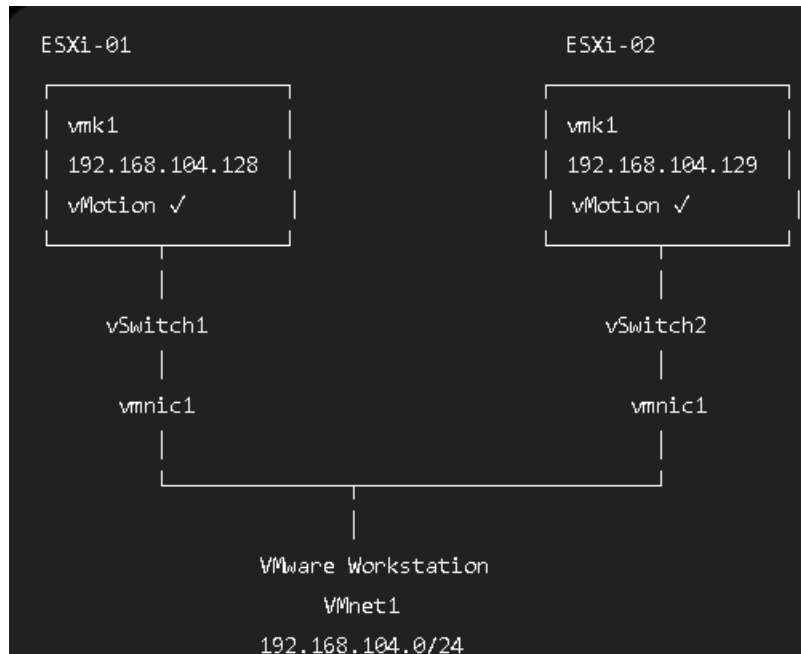


Cấu hình VmKernel Vmotion

Mô hình test



The screenshot shows the vSphere Client interface for two hosts. The top host (192.168.174.132) has two VMkernel adapters: vmk0 (Management Network, vSwitch0, IP 192.168.174.132) and vmk1 (vMotion, vSwitch1, IP 192.168.104.128). The bottom host (192.168.174.135) has two VMkernel adapters: vmk0 (Management Network, vSwitch0, IP 192.168.174.135) and vmk1 (vMotion, vSwitch2, IP 192.168.104.129). Both hosts have vMotion enabled for their respective vmk1 adapters.

Device	Network Label	Switch	IP Address	TCP/IP Stack	Enabled Services
vmk0	Management Network	vSwitch0	192.168.174.132	Default	vMotion +1
vmk1	vMotion	vSwitch1	192.168.104.128	Default	vMotion

Device	Network Label	Switch	IP Address	TCP/IP Stack	Enabled Services
vmk0	Management Network	vSwitch0	192.168.174.135	Default	vMotion +1
vmk1	vMotion	vSwitch2	192.168.104.129	Default	vMotion

Đảm bảo vmotion gắn vào đúng physical adapter (cái này phải ở trạng thái active, nếu không thì sẽ không cấp DHCP cho vmk1)

groups

▼ Physical Adapters

No physical network adapters

174.135

ACTIONS

Manage Physical Network Adapters | vSwitch2

✕

Select unclaimed (free) adapters on the host and move them to the Active/Standby/Unused adapters of the switch.

MOVE UP MOVE DOWN

Unclaimed adapters

vmnic1

Active adapters

Standby adapters

Unused adapters

▼ Properties

Adapter	VMware Inc. vmxnet3 Virtual Ethernet Controller
Name	vmnic1
Location	PCI 0000:13:00.0
Driver	vmxnet3

▼ Status

Status	Connected
Actual speed, Duplex	10 Gbit/s, Full Duplex
Configured speed, Duplex	10 Gbit/s, Full Duplex
Networks	0.0.0.1-255.255.255.254

▼ SR-IOV

Status	Not supported
--------	---------------

▼ Cisco Discovery Protocol

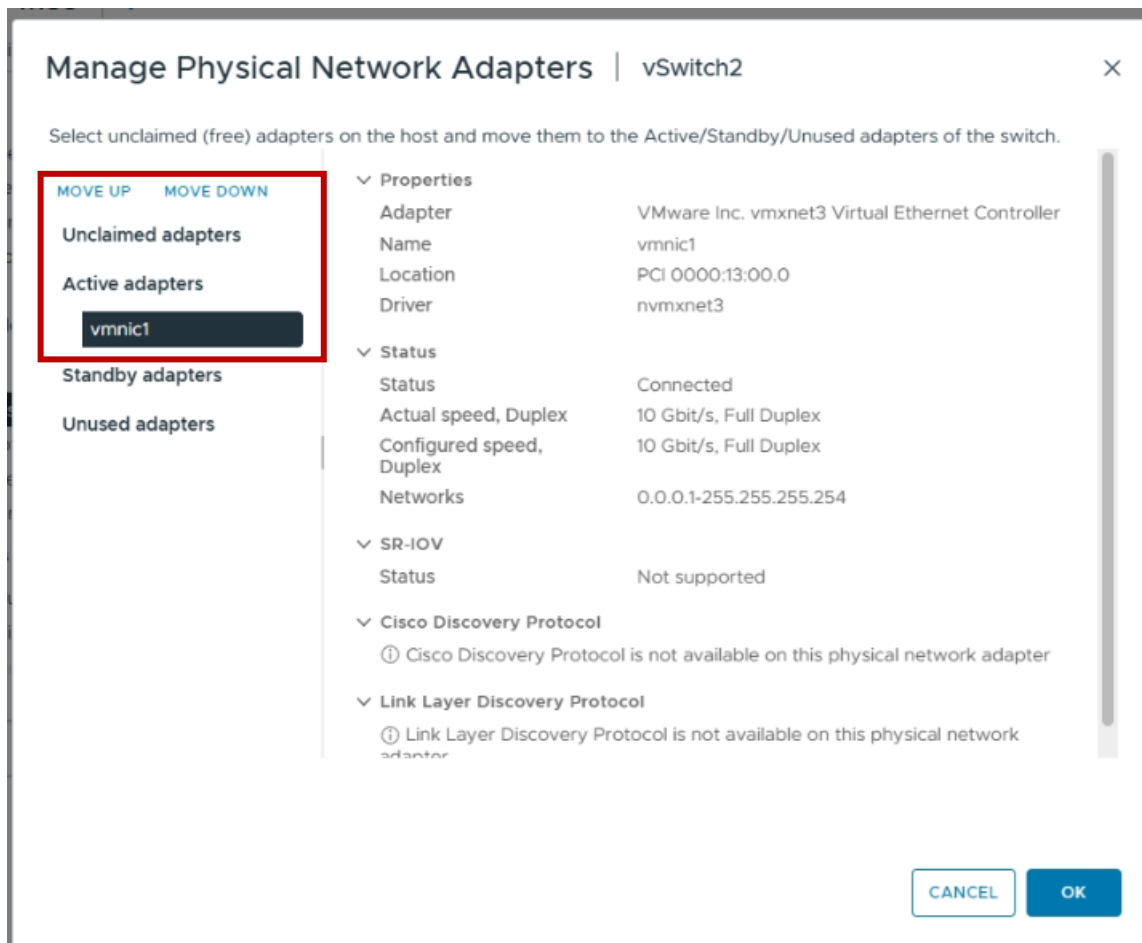
① Cisco Discovery Protocol is not available on this physical network adapter

▼ Link Layer Discovery Protocol

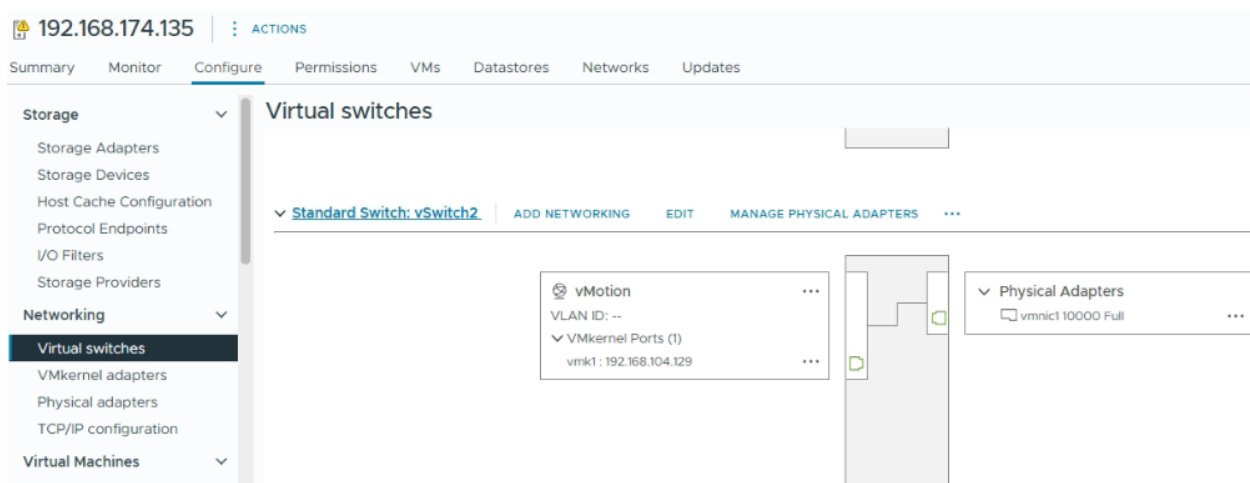
① Link Layer Discovery Protocol is not available on this physical network adapter

CANCEL

OK



Cấu hình chuẩn:



Test connection của 2 vmotion

Esxi 1: 192.168.174.132 ping tới vmk1 của Exsi2: 192.168.104.129

```
Windows PowerShell x Windows PowerShell x + v
[root@localhost:~] vmkping -I vmk1 192.168.104.129
PING 192.168.104.129 (192.168.104.129): 56 data bytes
64 bytes from 192.168.104.129: icmp_seq=0 ttl=64 time=16.984 ms
64 bytes from 192.168.104.129: icmp_seq=1 ttl=64 time=0.221 ms
64 bytes from 192.168.104.129: icmp_seq=2 ttl=64 time=0.476 ms

--- 192.168.104.129 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.221/5.894/16.984 ms

[root@localhost:~] |
```

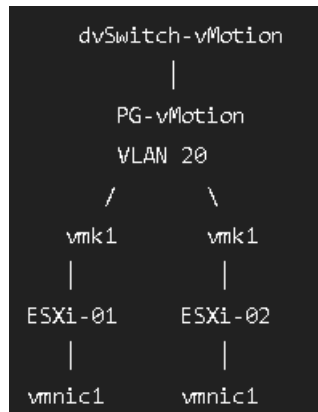
Esxi 2: 192.168.174.135 ping tới vmk1 của Exsi1: 192.168.104.128

```
[root@localhost:~] vmkping -I vmk1 192.168.104.128
PING 192.168.104.128 (192.168.104.128): 56 data bytes
64 bytes from 192.168.104.128: icmp_seq=0 ttl=64 time=0.411 ms
64 bytes from 192.168.104.128: icmp_seq=1 ttl=64 time=0.695 ms
64 bytes from 192.168.104.128: icmp_seq=2 ttl=64 time=0.855 ms

--- 192.168.104.128 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.411/0.654/0.855 ms

[root@localhost:~]
```

Distributed Switch



Datacenter1

ACTIONS

Summary

Monitor

Configure

Permissions

Hosts & Clusters

VMs

Datastores

Networks

Updates

Networks

Distributed Switches

Distributed Port Groups

Uplink Port Groups

Network Folders

Quick Filter

Enter value

☒	Name	Distributed Switch	VLAN ID	NSX Port Group ID	VNI	Port Binding
☒	PG-vMotion	dvSwitch-vMotion	VLAN access: 20			Static binding (elastic)

Networking

→ dvSwitch-vMotion

→ PG-vMotion

→ Edit Settings

→ VLAN type: VLAN

→ VLAN ID: 20

Test migrate: trên VM test vMotion

Mô hình test

```
BEFORE
VM-vMotion-Test
Host: ESXi-02 - 192.168.174.135
|
| vMotion
| vmk1 .129 ↔ .128
▼

AFTER
Host: ESXi-01 - 192.168.174.132

VM Powered On      ✓
IP không đổi      ✓
Không reboot      ✓
Datastore_NFS     ✓
Migration Completed ✓
```

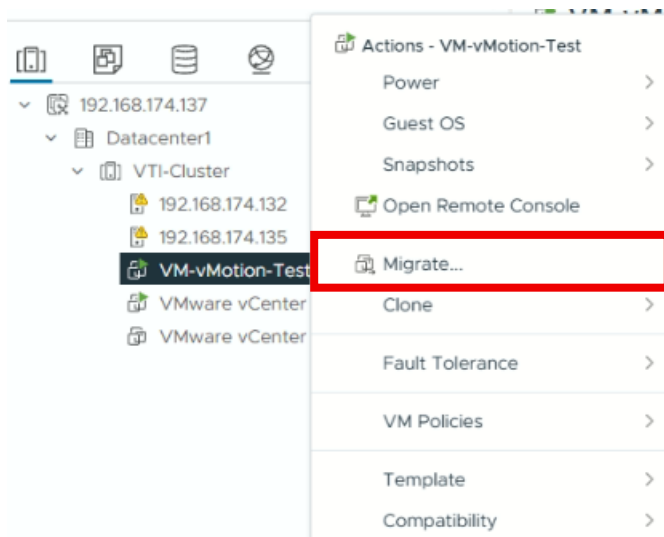
Kiểm tra connect giữa máy test tới gateway

```
vmotion@vmotion:~$ ping 192.168.174.2
PING 192.168.174.2 (192.168.174.2) 56(84) bytes of data.
64 bytes from 192.168.174.2: icmp_seq=1 ttl=128 time=0.651 ms
64 bytes from 192.168.174.2: icmp_seq=2 ttl=128 time=0.131 ms
64 bytes from 192.168.174.2: icmp_seq=3 ttl=128 time=0.192 ms
64 bytes from 192.168.174.2: icmp_seq=4 ttl=128 time=0.186 ms
64 bytes from 192.168.174.2: icmp_seq=5 ttl=128 time=0.164 ms
64 bytes from 192.168.174.2: icmp_seq=6 ttl=128 time=1.62 ms
64 bytes from 192.168.174.2: icmp_seq=7 ttl=128 time=0.186 ms
64 bytes from 192.168.174.2: icmp_seq=8 ttl=128 time=0.371 ms
64 bytes from 192.168.174.2: icmp_seq=9 ttl=128 time=0.513 ms
64 bytes from 192.168.174.2: icmp_seq=10 ttl=128 time=0.319 ms
^C
--- 192.168.174.2 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9242ms
rtt min/avg/max/mdev = 0.131/0.433/1.622/0.427 ms
vmotion@vmotion:~$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=128 time=44.8 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=128 time=45.5 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=128 time=44.3 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=128 time=44.9 ms
--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 44.331/44.878/45.514/0.420 ms
vmotion@vmotion:~$
```

ping liên tục máy test 192.168.174.139

```
Reply from 192.168.174.139: bytes=32 time<1ms TTL=64
Reply from 192.168.174.139: bytes=32 time<1ms TTL=64
Reply from 192.168.174.139: bytes=32 time<1ms TTL=64
Reply from 192.168.174.139: bytes=32 time<1ms TTL=64
Reply from 192.168.174.139: bytes=32 time<1ms TTL=64
Reply from 192.168.174.139: bytes=32 time<1ms TTL=64
Reply from 192.168.174.139: bytes=32 time<1ms TTL=64
Reply from 192.168.174.139: bytes=32 time<1ms TTL=64
Reply from 192.168.174.139: bytes=32 time<1ms TTL=64
Reply from 192.168.174.139: bytes=32 time<1ms TTL=64
Reply from 192.168.174.139: bytes=32 time<1ms TTL=64
Reply from 192.168.174.139: bytes=32 time<1ms TTL=64
Reply from 192.168.174.139: bytes=32 time<1ms TTL=64
Reply from 192.168.174.139: bytes=32 time<1ms TTL=64
Reply from 192.168.174.139: bytes=32 time=2ms TTL=64
```

Migrate vMotion trên máy test



Migrate | VM-vMotion-Test

- 1 Select a migration type
- 2 Select a compute resource
- 3 Select networks
- 4 Select vMotion priority
- 5 Ready to complete

Select a migration type

Change the virtual machines' compute resource, storage, or both.

VM ORIGIN ⓘ

☒ Change compute resource only

Migrate the virtual machines to another host or cluster.

☐ Change storage only

Migrate the virtual machines' storage to a compatible datastore or datastore cluster.

☐ Change both compute resource and storage

Migrate the virtual machines to a specific host or cluster and their storage to a specific datastore or datastore cluster.

☐ Cross vCenter Server export

Migrate the virtual machines to a vCenter Server not linked to the current SSO domain.

CANCEL

NEXT

Migrate | VM-vMotion-Test

- 1 Select a migration type
- 2 Select a compute resource
- 3 Select networks
- 4 Select vMotion priority
- 5 Ready to complete

Select a compute resource

Select a cluster, host, vApp or resource pool to run the virtual machines.

VM ORIGIN ⓘ

☐ Show all compute resources

Hosts Clusters Resource Pools vApps

Quick Filter

Enter value

	Name	State	Status	Cluster	Cons
☒	 192.168.174.132	Connected	 Warning	 VTI-Cluster	
☐	 192.168.174.135	Connected	 Warning	 VTI-Cluster	

Manage Columns2 items

Compatibility

✓ Compatibility checks succeeded.

CANCEL

BACK

NEXT

Migrate | VM-vMotion-Test

- 1 Select a migration type
- 2 Select a compute resource
- 3 **Select networks**
- 4 Select vMotion priority
- 5 Ready to complete

Select networks

Select destination networks for the virtual machine migration.

VM ORIGIN ⓘ

Migrate VM networking by selecting a new destination network for all VM network adapters attached to the same source network.

[illegible]

[ADVANCED >>](#)

Compatibility

✓ Compatibility checks succeeded.

CANCEL

[BACK](#)

NEXT

Migrate | VM-vMotion-Test

- 1 Select a migration type
- 2 Select a compute resource
- 3 Select networks
- 4 Select vMotion priority**
- 5 Ready to complete

Select vMotion priority

Protect the performance of your running virtual machines by prioritizing the allocation of CPU resources.

VM ORIGIN ⓘ

- ☒ **Schedule vMotion with high priority (recommended)**
vMotion receives higher CPU scheduling preference relative to normal priority migrations. vMotion might complete more quickly.
- ☐ **Schedule normal vMotion**
vMotion receives lower CPU scheduling preference relative to high priority migrations. You can extend vMotion duration.

CANCEL

[BACK](#)

NEXT

Migrate | VM-vMotion-Test

- 1 Select a migration type
- 2 Select a compute resource
- 3 Select networks
- 4 Select vMotion priority
- 5 Ready to complete

Ready to complete

Verify that the information is correct and click Finish to start the migration.

VM ORIGIN ⓘ

Migration Type	Change compute resource. Leave VM on the original storage
Virtual Machine	VM-vMotion-Test
Cluster	VTI-Cluster
Host	192.168.174.132
vMotion Priority	High
Networks	No network reassignments

CANCEL

BACK

FINISH

Migrate thành công qua máy exsi 1

VM-vMotion-Test

Summary

Monitor

Configure

Permissions

Datastores

Networks

Snapshots

Updates

Guest OS



LAUNCH REMOTE CONSOLE ⓘ

LAUNCH WEB CONSOLE

Virtual Machine Details

Power Status

Powered On

Guest OS

Ubuntu Linux (64-bit)

VMware Tools

Running, version:12448 (Guest Managed) ⓘ

DNS Name

IP Addresses

Encryption

Not encrypted

Usage

Last updated: 9/8/26, 1:47 PM

CPU

24 MHz used

Memory

61 MB used

Storage

4.2 GB used

VIEW STATS

VM Hardware

CPU

2 CPU(s), 24 MHz used

Memory

2 GB, 0 GB memory active

Hard disk 1

25 GB | Thin Provision ⓘ
Datastore_NFS

Network

VM Network (connected) | 00:50:56:89-f:29

PCI Devices

ⓘ

Related Objects

Cluster

VTI-Cluster

Host

192.168.174.132

Networks

VM Network

Máy ping liên tục trong quá trình migrate và uptime full không bị gián đoạn => VM **không reboot** trong quá trình vMotion

```
Reply from 192.168.174.139: bytes=32 time<1ms TTL=64
Reply from 192.168.174.139: bytes=32 time<1ms TTL=64
Reply from 192.168.174.139: bytes=32 time<1ms TTL=64
Reply from 192.168.174.139: bytes=32 time<1ms TTL=64
Reply from 192.168.174.139: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.174.139:
    Packets: Sent = 1202, Received = 1202, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 6ms, Average = 0ms
Control-C
PS C:\Users\hung>
```

```
06:50:49 up 22 min,  1 user,  load average: 0.00, 0.00, 0.00
vmotion@vmotion:~$ uptime
06:50:56 up 22 min,  1 user,  load average: 0.00, 0.00, 0.00
vmotion@vmotion:~$
```

Test migrate thành công!

! Lỗi báo khi migrate trên vlan 20

The vMotion migrations failed because the ESX hosts were not able to connect over the vMotion network. Check the vMotion network settings and physical network configuration.

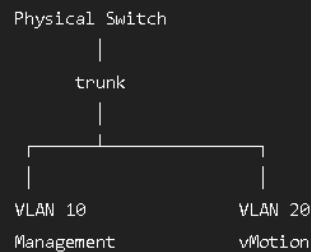
2026-09-08T04:24:23.525026Z vMotion migration [-1062687097:2676370995435219693] the remote host closed the connection unexpectedly and migration has stopped. The closed connection probably results from a migration failure detected on the remote host. Failed waiting for data. Error 195887167. Connection closed by remote host, possibly due to timeout.

2026-09-08T04:24:24.253652Z Migration [-1062687097:2676370995435219693] failed to connect to remote host <192.168.104.128> from host <192.168.104.129>: Host is down.

VMotionStream [-1062687097:2676370995435219693] stream thread failed to connect to the remote host <192.168.104.128>: The ESX hosts failed to connect over the VMotion network

Cho nên về mặt **network isolation**, bạn vẫn đang tách Management và vMotion thành hai broadcast domain khác nhau.

Trong production người ta thường làm:



Còn trong Workstation lab của bạn đang mô phỏng bằng:



Nói cách khác:

VMnet1 đang thay thế vai trò phân đoạn mạng mà VLAN 20 thường đảm nhiệm trong hạ tầng thật.

Cách khắc phục => để vtag Vlan về none thay vì đặt vlan 20

Giải thích

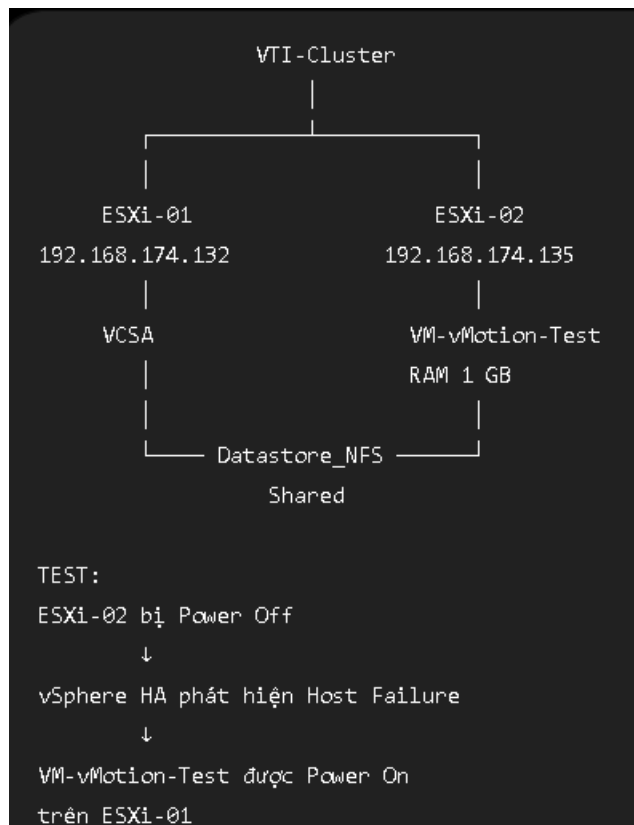
Trong môi trường Nested ESXi trên VMware Workstation, mạng vMotion được cô lập bằng VMnet1 riêng.

Khi cấu hình VLAN ID 20 trên Distributed Port Group, ESXi gắn thẻ 802.1Q VLAN 20 vào lưu lượng vMotion, trong khi mạng VMnet1 bên dưới không được triển khai như một trunk VLAN 20 end-to-end, dẫn đến lỗi kết nối vMotion.

Khi đặt VLAN type về None, lưu lượng được truyền untagged trên VMnet1 và vMotion hoạt động thành công.

HA vSphere

Mô hình test



Cấu hình vSphere HA

Edit Cluster Settings | VTI-Cluster

vSphere HA

Failures and responses

Admission Control

Heartbeat Datastores

Advanced Options

You can configure how vSphere HA responds to the failure conditions on this cluster. The following failure conditions are supported: host, host isolation, VM component protection (datastore with PDL and APD), VM and application.

Enable Host Monitoring

> Host Failure Response	Restart VMs
> Response for Host Isolation	Disabled
> Datastore with PDL	Power off and restart VMs
> Datastore with APD	Power off and restart VMs - Conservative restart policy
> VM Monitoring	Disabled

Test failures: máy vMotion đang trên esxi 2

Related Objects

Cluster

[VTI-Cluster](#)

Host

[192.168.174.135](#)

Networks

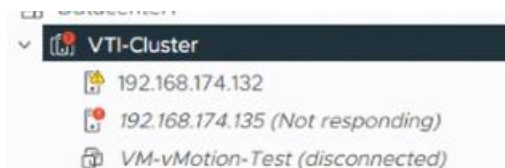
[VM Network](#)

Shutdown ESXi đội ngọt giả lập mất điện



Máy vMotion mất kết nối

```
Reply from 192.168.174.139: bytes=32 time<1ms TTL=64
Reply from 192.168.174.139: bytes=32 time<1ms TTL=64
Reply from 192.168.174.139: bytes=32 time=1ms TTL=64
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
```



Đợi máy vMotion reboot lại

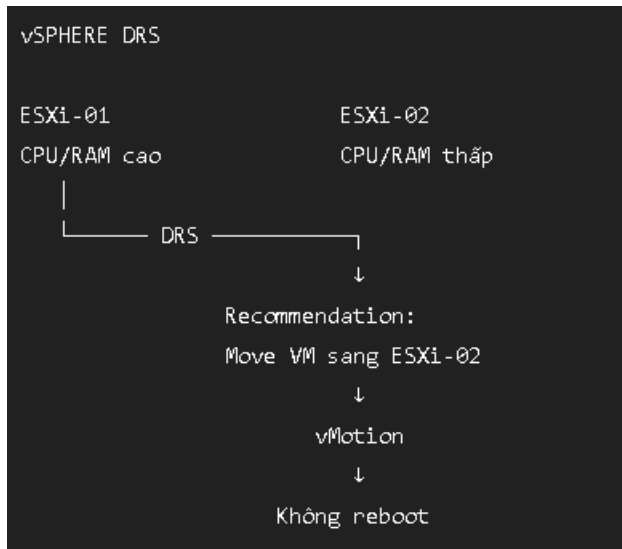
```
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Reply from 192.168.174.139: bytes=32 time=1ms TTL=64
Reply from 192.168.174.139: bytes=32 time<1ms TTL=64
Reply from 192.168.174.139: bytes=32 time<1ms TTL=64
Reply from 192.168.174.139: bytes=32 time<1ms TTL=64
Reply from 192.168.174.139: bytes=32 time=1ms TTL=64
```

Máy vMotion đã tự động migrated qua ESXi 1 thành công



vSPHERE DRS Manual Vmotion

mô hình test



vMotion hiện đang trên ESXi 1

Test bằng cách stress cpu trên Vmotion làm quá tải trên EXSi 1 → DRS sẽ migrated máy vMotion qua EXSi 2

Kiểm tra cấu DRS Score

VTI-Cluster

ACTIONS

Summary

Monitor

Configure

Permissions

Hosts

VMs

Datastores

Networks

Updates

Issues and Alarms

All Issues

Triggered Alarms

Performance

Overview

Advanced

Tasks and Events

Tasks

Events

vSphere DRS

Recommendations

Faults

History

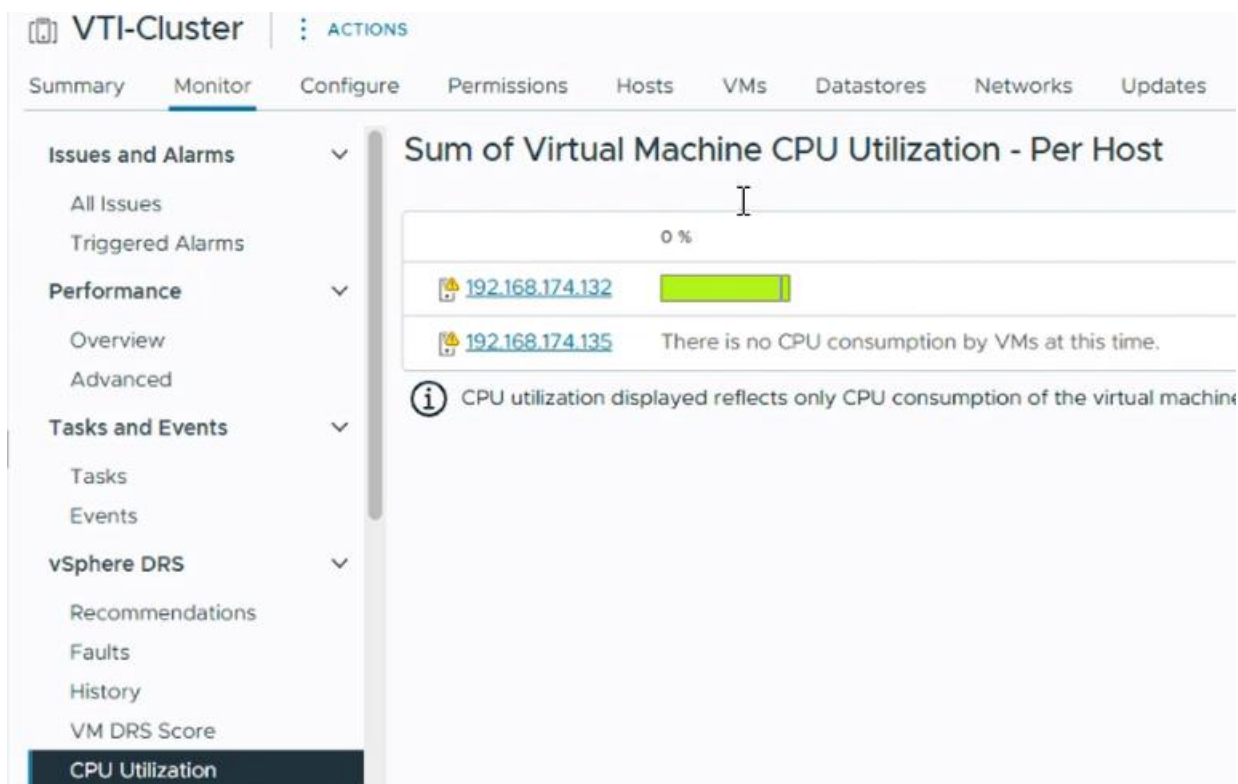
VM DRS Score

VM DRS Score

Quick Filter

Enter value

☐	Name	DRS Score	Active CPU	Used CPU	CPU Readiness	Granted Memory
☐	vCLS-332a4d56-d807-dab2-2ea7-81dbfe12258f	--	0 Hz	0 Hz	0%	130 MB
☐	vCLS-55c54d56-5303-27fe-9837-85a874fe827e	--	0 Hz	0 Hz	0%	128 MB
☐	VMware vCenter Server (1)	--	0 Hz	0 Hz	--	0 B
☐	VMware vCenter Server	67%	648 MHz	599 MHz	2%	13.98 GB
☐	VM-vMotion-Test	72%	474 MHz	449 MHz	1%	1,020 MB



Ping vMotion liên tục để kiểm tra

```
PS C:\Users\hung> ping -t 192.168.174.139

Pinging 192.168.174.139 with 32 bytes of data:
Reply from 192.168.174.139: bytes=32 time<1ms TTL=64
Reply from 192.168.174.139: bytes=32 time<1ms TTL=64
```

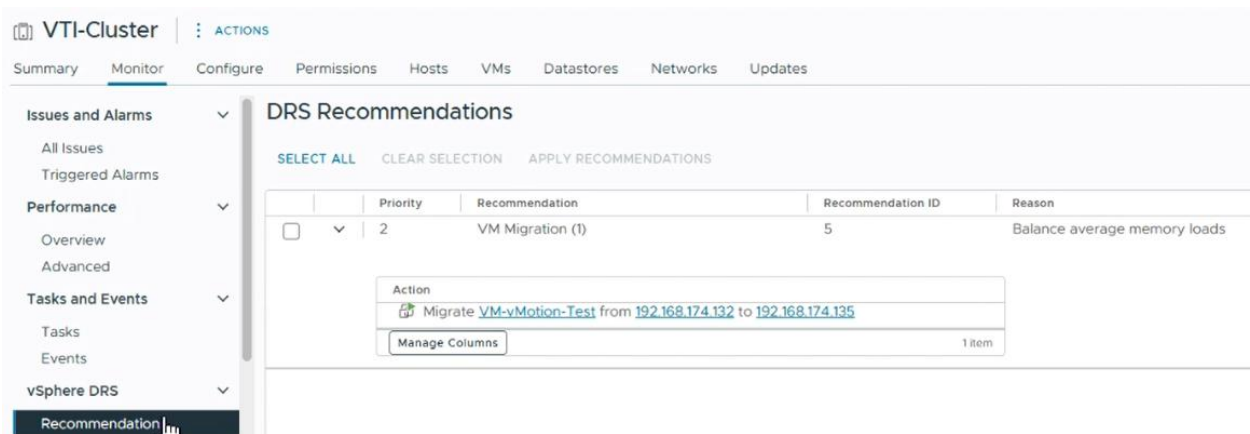
Stress cpu trên vMotion

```
vmotion@vmotion:~$ stress-ng --cpu 2 --cpu-load 100 --timeout 30m
stress-ng: info: [2164] setting to a 30 mins, 0 secs run per stressor
stress-ng: info: [2164] dispatching hogs: 2 cpu
stress-ng: info: [2165] cpu: for stable load results, select a specific cpu stress method with 'all'
stress-ng: info: [2166] cpu: for stable load results, select a specific cpu stress method with 'all'
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
2165	vmotion	20	0	64864	5748	2632	R	100.0	0.6	0:46.52	stress-ng-cpu
2166	vmotion	20	0	64864	5748	2632	R	100.0	0.6	0:46.76	stress-ng-cpu
2233	vmotion	20	0	11916	5696	3492	R	10.0	0.6	0:00.01	top

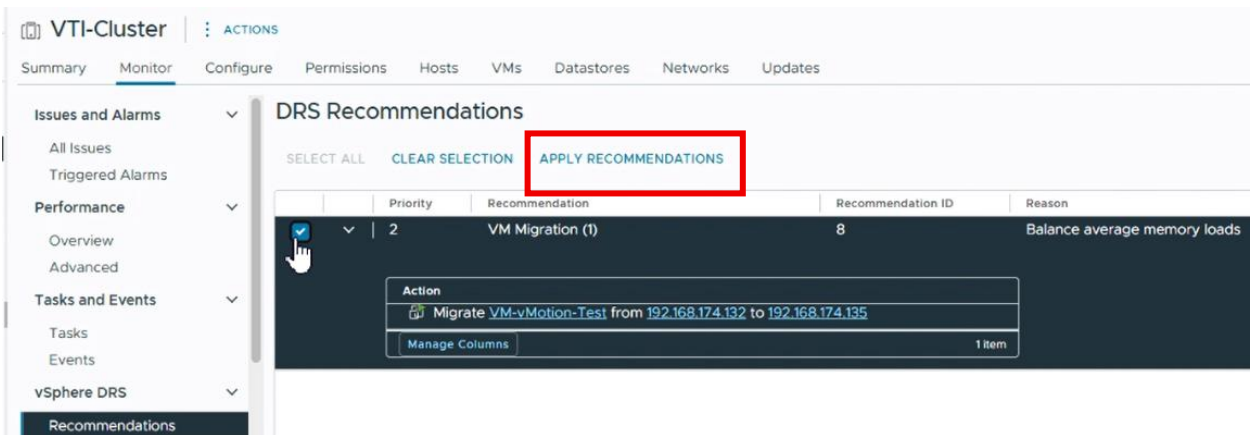
Kiểm tra DRS Recommendation

VTI-Cluster → Monitor → vSphere DRS → Recommendations



Action: migrated vMotion sang ESXi 2

Apply Recommendations



Sau khi apply hệ thống migrated Vmotion sang ESXi 2

Recent Tasks		Alarms						
Task Name	Target	Status	Details	Initiator	Queued For	Start Time	Completion Time	Server
Migrate virtual machine	VM-vMotion-Test	84%	Migrating Virtual Machine active state	System	9 ms	09/09/2026, 2:45:28 PM		192.168.174.137

Trong quá trình Migrate máy Vmotion vẫn không reboot → test DRS thành công

The screenshot shows the vSphere Web Client interface for a cluster named 'VTI-Cluster'. The 'Monitor' tab is selected, and the 'vSphere DRS' section is expanded, showing 'Recommendations'. A table titled 'DRS Recommendations' is visible, with columns for 'Priority' and 'Recommendation'. Below the table, a status bar shows '84%' completion and 'Migrating Virtual Machine active state'. To the right, a terminal window displays a series of network traffic logs, including 'Reply from 192.168.174.139: bytes=32 time<1ms TTL=64'.

Đã migrate thành công sang ESXi 2

The screenshot shows the 'Related Objects' section of the vSphere Web Client. It lists several objects: 'Cluster' (VTI-Cluster), 'Host' (192.168.174.135), 'Networks' (VM Network), and 'Storage' (Datastore NFS). The 'Host' object is highlighted with a red box.

WARP cá nhân:

Laptop → WARP tunnel → Cloudflare → Internet

Zero Trust:

Laptop → WARP tunnel → Cloudflare

